

Fateme GHAYEM

2023–Now	Postdoctoral research fellow TANGRAM team, Inria, Nancy, France
CONTACT INFORMATION	Email: fatemeh.ghayem@inria.fr fateme.ghayem@gmail.com Homepage: https://ghayem.github.io ORCID: https://orcid.org/0000-0001-5779-6685
EDUCATION	Université Grenoble Alpes , Grenoble, France Ph.D. in Signal, Image, Parole, and Télécoms, GIPSA-lab, Oct 2017 – Nov 2020 - Thesis topic: <i>Optimal sensor placement for source extraction</i> - Advisor: Prof. Christian JUTTEN, Dr. Bertrand RIVET Sharif University of Technology , Tehran, Iran M.Sc., Electrical Engineering, Sept 2013 – Sept 2015 GPA: 17.06/20 - Thesis topic: <i>MR image reconstruction from highly partial Fourier samples</i> - Advisor: Prof. Farokh Marvasti Shiraz University , Shiraz, Iran B.Sc., Electrical Engineering, Sept 2009 – Sept 2013 GPA: 17.85/20 National Organization for Development of Exceptional Talents , Shiraz, Iran Diploma, Mathematics and Physics, Sept 2005 – Sept 2009 GPA: 19.60/20
RESEARCH INTERESTS	– Image/text multi-modal foundation models for brain study – Brain meta-analysis with large language models (LLM) – Independent component/vector analysis (ICA/IVA) for multi-subject fMRI study – Aneurysm detection with deep learning – Optimal sensor placement for source extraction – Machine learning and statistical signal processing in medical applications – Bayesian modeling, numerical optimization, and dictionary learning
RESEARCH EXPERIENCES	• Postdoctoral researcher (2025–Now), TANGRAM, Inria Nancy, France. - Advisor: Dr. Erwan Kerrien - Research topic: Deep learning methods for aneurysm detection. - Summary: The project focuses on developing deep learning approaches for reliable aneurysm detection from MRA and related modalities. The objectives are to: * Improve accuracy and robustness of aneurysm detection using advanced neural architectures; * Improve weakly annotated datasets for model training, as well as apply data augmentation techniques and multi-modal data analysis to address the data size limitation.

- **Postdoctoral researcher** (March 2023–2025), MIND, Inria Paris-Saclay, France.
 - Advisor: [Dr. Bertrand Thirion](#)
 - Co-advisor: [Dr. Demian Wassermann](#)
 - Research topic: Knowledge and representation integration on the brain
 - Summary:

The project aims to develop a novel approach for image representation in the context of brain imaging, enabling a more nuanced and flexible assessment of the associations between images and arbitrary queries, unconstrained by traditional “bag of words” limitations. In other words, we want to:

 - * Provide reliable knowledge from diverse brain studies;
 - * Address challenges such as lack of statistical power in individual studies, reproducibility, and terminology inconsistency;
 - * Provide insights into the relationship between brain structure and behavior.
- **Postdoctoral researcher** (August 2021–August 2022), [MLSP-Lab](#), University of Maryland, Baltimore County (UMBC), Maryland, USA.
 - Advisor: [Prof. Tulay Adali](#)
 - Research topics:
 - Dictionary learning for the identification of new interpretable patterns and discriminative features from brain functional network connectivity (FNC) obtained from ICA decomposition of multi-subject resting-state fMRI data for static and dynamic studies.
 - Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, *e.g.*, Schizophrenia.
 - Constrained ICA and IVA for subgroup identification from multisubject fMRI Data.
 - Reproducibility and replicability in neuroimaging data analysis.
- **Research assistant** (2015–2017), DSP-lab, EE Department, Sharif University of Technology, Tehran, Iran.
 - Advisor: [Prof. Massoud Babaie-Zadeh](#)
 - Research topics: Dictionary learning for sparse representation, convex/non-convex optimization.

POSTERS “NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature”, [Organization for Human Brain Mapping \(OHBM\)](#), Brisbane, Australia, 2025.

WORKSHOP [COGBASES workshop on open science methods for analyzing brain imaging data](#), Paris, France, October 2023.

SUMMER SCHOOL [PRAIRIE artificial intelligence summer school \(PAISS\)](#), Grenoble, France, July 2018.

TALKS

- **From Contrastive Text-to-Brain Mapping to Multiview Learning of Depressive Disorders through Biological and Discourse Biomarkers**, [MULTISPEECH](#), Inria-Nancy, France, February 2026.
- **Multiview Learning of the Brain: Functional Connectivity, Subgroup Discovery, and Text-to-Brain Mapping**, *IMoPA*, *ERCN*, *CHRU de Nancy*, Nancy, France, February 2026.

- **NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation**, *The British Council in Paris*, Paris, France, September 2025.
- **Exploring Brain Function and Structure: From Sparse Coding to Multimodal Meta-Analysis**, *Laboratoire de Physique de l'ENS de Lyon*, Lyon, France, February 2024.
- **Multi-subject fMRI Analysis for Brain Pattern Discovery and Subgroup Identification**, *MIND team, Inria-Saclay*, Paris, France, June 2024.
- **New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity using Dictionary Learning**, *MIND team, Inria-Saclay*, Paris, France, September 2023.
- **Optimal Sensor Placement for Source Extraction**, *MIND team, Inria-Saclay*, Paris, France, June 2023.
- **Optimal Sensor Placement for Source Extraction**, *Diagnostic and Interventional Adaptive Imaging (IADI)*, Nancy, France, January 2023.
- **Optimal Sensor Placement for Source Extraction**, *Centre de Recherche en Automatique de Nancy (CRAN), Department of Biology, Signals and Systems*, Nancy, France, January 2021.

Publications

PREPRINTS

1. **F. Ghayem**, R. Meudec, J. Dockès, B. Thirion, D. Wassermann, “NeuroConText: Contrastive Learning for Neuroscience Meta-Analysis with Rich Text Representation,” submitted to *Imaging Neuroscience*, July 2025. [\[Paper\]](#) [\[Codes\]](#)
2. R. Meudec, J. Dockès, **F. Ghayem**, D. Wassermann, B. Thirion, “Peaks2Image: Reconstructing fMRI Maps from Stereotactic Coordinates to Enhance Cognitive Meta-Analysis,” submitted to *Imaging Neuroscience*, August 2025. [\[Paper\]](#)

REFEREED JOURNAL PUBLICATIONS

1. **F. Ghayem**, B. Rivet, C. Jutten, R. Cabral Farias, “Robust sensor placement for signal extraction”, *IEEE Transactions on Signal Processing*, vol. 69, pp. 4513-4528, 2021.
2. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, “Sparse signal recovery using iterative proximal projection”, *IEEE Transactions on Signal Processing*, vol. 66, no. 4, pp. 879–894, February 2018.

CONFERENCE PUBLICATIONS

1. R. Meudec*, **F. Ghayem***, J. Dockès, D. Wassermann, B. Thirion, “NeuroConText: Contrastive Text-to-Brain Mapping for Neuroscientific Literature”, *International Conference on Medical Image Computing And Computer Assisted Intervention (MICCAI)*, October 2024.
* [Equal contribution](#)
2. **F. Ghayem**, H. Yang, F. Kantar, S-J. Kim, V. D. Calhoun, T. Adali, “New Interpretable Patterns and Discriminative Features from Brain Functional Network Connectivity Using Dictionary Learning”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Rhodes island, June 2023.

3. H. Yang, **F. Ghayem**, B. Gabrielson, M. A. B. S. Akhonda, V. D. Calhoun, T. Adali, “Constrained independent component analysis based on entropy bound minimization for subgroup identification from multisubject fMRI data”, *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, June 2023.
4. H. Yang, MABS. Akhonda, **F. Ghayem**, Q. Long, VD. Calhoun, T Adali, “Independent Vector Analysis Based Subgroup Identification from Multisubject fMRI Data”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2022.
5. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, “Gradient-based algorithm with spatial regularization for optimal sensor placement”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2020.
6. **F. Ghayem**, B. Rivet, Ch. Jutten, R. Cabral Farias, “Optimal sensor placement for signal extraction”, in *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, May 2019.
7. M. Sadeghi, **F. Ghayem**, M. Babaie-Zadeh, S. Chatterjee, M. Skoglund, and C. Jutten, “L0Soft: ℓ_0 Minimization via Soft Thresholding”, in *Proceedings of the 27th European Signal Processing Conference (EUSIPCO)*, 2-6 September 2019.
8. **F. Ghayem**, M. Sadeghi, M. Babaie-Zadeh, and C. Jutten, “Accelerated dictionary learning for sparse signal representation”, in *13th International Conference on Latent Variable Analysis and Signal Separation, LVA/ICA*, Grenoble, France, 2017.
9. **F. Ghayem** and F. Rassaie, “Helical antenna to measure radiated power density around a BTS; Design and implementation”, in *third Asia-Pacific Conference on Antennas and Propagation (APCAP)*, July 2014.

CO-SUPERVISION I am currently supervising four Master 1 students at Université de Lorraine on a project titled “Enhanced NeuroConText with deep image representation for brain meta-analysis”. Also, I co-supervised three Ph.D. students in collaboration with Prof. Tulay Adali (University of Maryland, Baltimore County, USA), Prof. Jean-Christophe Pesquet (CentraleSupélec, Université Paris-Saclay, France), Prof. Vince D. Calhoun (Translational Research in Neuroimaging and Data Science, USA), and Dr. Seung-Jun Kim (University of Maryland, Baltimore County, USA) on the following projects:

- Brain graph neural networks (Brain-GNN) for the classification of healthy control and patients with different brain disorders, e.g. Schizophrenia.
- ICA and IVA for subgroup identification from multisubject resting state fMRI Data.
- Reproducibility and replicability in neuroimaging data analysis.

TEACHING	Computer Vision and Image Processing	2026
	- 20 hours practical sessions	
	- Polytech Nancy	
	Signal Processing	2025-2026
	- 6 hours lectures + 12 hours practical sessions	
	- Université de Lorraine	
	Signal Processing	2024-2025
	- 6 hours lectures + 12 hours practical sessions	
	- Université de Lorraine	
	Signals and Systems	2015
	- 10 hours practical sessions	

	- Responsible: Prof. M. Babaei-Zadeh, Sharif University of Technology	
	Digital Signal Proccessing II	2014
	- 10 hours practical sessions	
	- Responsible: Prof. F. Marvasti, Sharif University of Technology	
	Signals and Systems	2013
	- 20 hours practical sessions	
	- Responsible: Dr. M. Derakhtian, Shiraz University	
	Electromagnetics	2012
	- 20 hours practical sessions	
	- Responsible: Dr. M. Derakhtian, Shiraz University	
	Electrical Circuit II	2011
	- 20 hours practical sessions	
	- Responsible: Prof. M. A. Masnadi-Shirazi, Shiraz University	
QUALIFICATIONS	I am qualified to serve as Maître de Conférences, Section 26 – Mathématiques appliquées et applications des mathématiques.	
HONORS & AWARDS	- IEEE TMI Distinguished Reviewer Silver Level 2023 – 2024	2024
	- Ph.D. scholarship (ranked 2), Université Grenoble Alpes, Grenoble, France.	2017
	- Full travel grant (CHESS project), LVA/ICA workshop, Grenoble, France.	2017
	- Bronze award in math competition among high school students, Sharif University of Technology.	2008
	- Admitted to National Organization for Development of Exceptional Talents (NODET) as high school and pre-university school student.	2005
COMPUTER SKILLS	• <i>Programming Languages and Softwares:</i> Python, PyTorch, MATLAB • <i>Typesetting:</i> $\text{\LaTeX}$ • <i>Toolbox:</i> Nilearn, GIFT	
COMMUNITY SERVICES	Reviewer for the following journals and conferences:	
	- AAAI	2026
	- ICML	2025
	- NeurIPS	2024-2025
	- Journal of Machine Learning Research	2024-2025
	- IEEE Transactions on Machine Learning Research	2024-2025
	- IEEE Transactions on Medical Imaging	2023-2026
	- IEEE Transactions on Signal Processing	2019-2022
	- IEEE Signal Processing Letters	2019, 2021, 2023
	- International Conf. on Acoustics, Speech, and Signal Proc. (ICASSP)	2023
	- European Signal Processing Conference (EUSIPCO)	2019-2021
	- eNeuro	2022
	- Machines	2022-2023
LANGUAGE PROFICIENCY	- English (Fluent)	
	- French (Intermediate)	
	- Persian (Native)	
HOBBIES AND INTERESTS	Playing the violin, running, hiking, biking	

REFERENCES

- Prof. Christian Jutten
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